Comparative Analysis Report

Hybrid RAG & Fine-Tuning for Customer Support · Baseline vs Solution V1 vs Solution V2

1. Summary

We built three versions of the support bot and tested all of them on the same 390 test queries (27 intents, none of these queries were seen in train or valid). All three use the same model, `Qwen/Qwen2.5-1.5B-Instruct`, for writing the final answer.

- **Baseline** - just the model, no company info given to it.
- **V1 (Naive RAG)** - we embed the customer message as it is (all-MiniLM-L6-v2), pull the closest SOP from ChromaDB and put it in the prompt.
- **V2 (Hybrid RAG)** - a LoRA fine-tuned copy of the model first reads the message and gives back `{"intent", "category"}` JSON. We use the intent to look up the right SOP and then answer from that.

Short version of what we found: the fine-tuned router is where most of the gain came from. Intent accuracy went from 56.4% with the base model to 96.4% after fine-tuning, and on the adversarial queries it went from 50% to 97.4%. Because of that, retrieval Hit@1 jumped from 44.1% in V1 to 99.0% in V2. ROUGE-L also went up at every step (0.065, then 0.079, then 0.085), so V2 is about 31% better than baseline on that. Hallucinations came down from 4.1% to 3.1% once we added the SOP, but V2 did not bring them down any further than V1 did.

So the router and the retrieval part are basically working now. The weak part is the 1.5B generator, which still doesn't always stick to the SOP even when it has the right one in front of it.

2. How we evaluated

Item	What we used
Test data	390 rows, from a stratified 80/10/10 split of the 3,896 cleaned rows. 0 overlap with train/valid.
Adversarial subset	38 test rows we filtered with a regex for angry, rude or unsure wording (ugh, bloody, STILL, wanna, seriously etc.)
ROUGE/BLEU reference	Full text of the SOP that the true intent maps to (13 SOPs, around 230-280 words each)
V2 search string	Full text of the mapped SOP. Each SOP is one chunk so every intent finds its own SOP (27/27)
Decoding	Greedy. Same gen settings, same generator and same RAG prompt for all three
Hardware	Colab T4, torch 2.11, transformers 5.18, peft 0.21

One thing to keep in mind is that the ROUGE and BLEU numbers are low in general. That's because we compare a 2-3 line answer with a full policy document, so these scores only make sense when comparing the three systems with each other. Higher just means the answer used more of the right policy's wording. It doesn't tell you if the answer actually reads well.

3. Results

Metric	Baseline	V1 Naive RAG	V2 Hybrid RAG	V1 vs Base	V2 vs V1	V2 vs Base
Router format adherence (%)	-	-	100.0			
Intent accuracy, test (%)	-	-	96.41			
Intent accuracy, adversarial (%)	-	-	97.37			
Retrieval Hit@1 (%)	-	44.10	98.97		+124.4%	
Response format adherence (%)	100.0	100.0	100.0	0.0%	0.0%	0.0%
ROUGE-1	0.1182	0.1226	0.1263	+3.8%	+3.0%	+6.9%
ROUGE-L	0.0649	0.0789	0.0849	+21.5%	+7.6%	+30.8%
BLEU	0.0005	0.0025	0.0032	+372.8%	+28.9%	+509.6%
Consistency rate (%)	100.0	100.0	100.0	0.0%	0.0%	0.0%
Hallucination frequency (%)	4.10	3.08	3.08	-25.0%	0.0%	-25.0%

From `Comparative_Results_Summary.csv` in the Solution V2 notebook. The BLEU % changes look huge only because the baseline BLEU is almost zero.

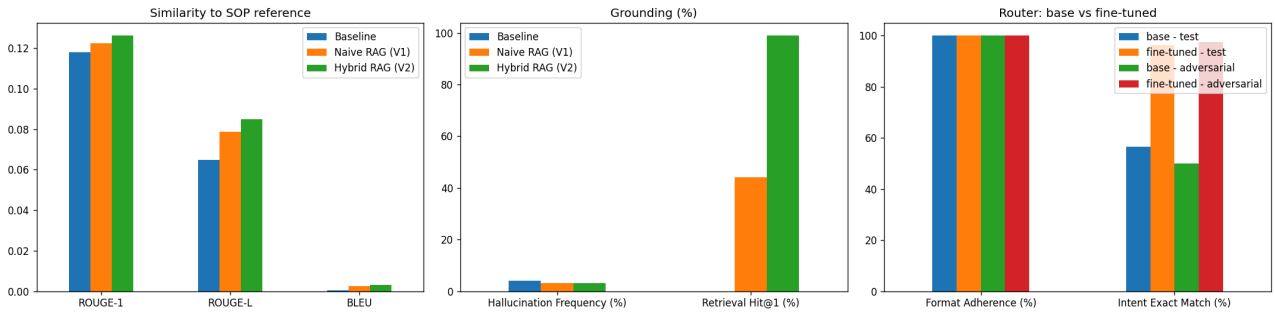


Figure 1. Left: similarity to the SOP. Middle: hallucination rate and Hit@1. Right: base router vs fine-tuned router.

Router - base vs fine-tuned (same system prompt for both)

Metric	Base, test	Fine-tuned, test	Base, adversarial	Fine-tuned, adversarial
Format adherence (%)	100.0	100.0	100.0	100.0
Intent exact match (%)	56.41	96.41	50.00	97.37
Intent + category exact (%)	50.77	96.41	44.74	97.37
Intent fuzzy match (%)	57.95	96.41	50.00	97.37

How we did against the targets in our proposal

Metric	Target	What we got	Status
Router format adherence	95% or more	100%	**Met**
Intent accuracy	85%+ on test, 80%+ on adversarial	96.4% / 97.4%	**Met**
Retrieval Hit@1 (V2)	90%+ and clearly above V1	99.0% vs 44.1%	**Met**
ROUGE / BLEU order	V2 > V1 > Baseline	Holds for all three	**Met**
Consistency	100%	100% for all	**Met**
Hallucination frequency	At least 50% lower from Baseline to V2	25% lower (4.10% to 3.08%)	**Not met**

4. What each stage added

4.1 Baseline to V1 (adding retrieval)

Just giving the model some SOP improved ROUGE-L by 21.5% and the flagged hallucinations went down from 16 to 12. Unsupported times/amounts stayed the same at 1.3%, placeholders and made-up contacts went from 2.6% to 1.8%, and asking the customer for secrets went from 0.3% to 0.

The baseline probes in notebook 3 show the problem pretty clearly. Without any context the model told a customer refunds come in "1-2 business days" and that a reset link is valid for "12 hours". Neither of these is in any of our SOPs. 2 out of the 5 probe answers got flagged.

The issue with V1 is the retriever. It got the right SOP for only 172 of 390 queries (44.1%). create_account, review, place_order and delivery_options were the worst, 15-16 misses each. Short and vague messages like "switch to premium account" just don't look like any policy text. We also saw that when V1 got the right SOP its ROUGE-L was 0.110, and when it got the wrong one it dropped to 0.054. So a wrong SOP is pretty much as useless as having none.

4.2 V1 to V2 (adding the fine-tuned router)

For the adapter we used LoRA with r=16 on `q_proj` and `v_proj`, which is about 2.18M trainable params (0.14% of the model). It learned fast. Val loss went from 0.100 down to 0.0067 by step 225, then it stopped improving and early stopping ended the run at step 300 (16.3 min on the T4). We kept the step 225 checkpoint.

Format wasn't really a problem to begin with, the base model already gave valid JSON 100% of the time. What fine-tuning fixed was picking the right intent. The base model was only right about half the time out of 27 intents.

With the router in place Hit@1 went up to 99.0%. Routing fixed the SOP for 214 queries and didn't break any query that V1 already had right. On those 214, ROUGE-L went from 0.055 to 0.066.

Over the whole test set though, V2 is only 7.6% better than V1 on ROUGE-L. That is a lot smaller than the jump in retrieval. Our guess is that the 1.5B model rewrites things in its own words and adds a lot of generic support lines no matter which SOP it gets, so a better SOP only partly shows up in the final answer.



Figure 2. Train vs val loss for the router adapter (val every 25 steps, patience 3).

4.3 Adversarial queries

On angry, sarcastic or unsure messages the base router dropped to 50%. The fine-tuned one got 97.4%, which is actually a bit higher than its 96.4% on the full test set. Bitext already has a lot of messy wording in it, so we think the adapter learned to ignore the tone and just look at what the customer is asking for. That was the main reason we wanted a fine-tuned router in the first place.

Our demo query was "*ugh my order STILL hasn't arrived... how many days is delivery even supposed to take??*". V2 routed it to `delivery_period` and pulled `shipping_delays.md`, which is correct, and the answer didn't make up any contact details. But it quoted the international window (10-21 business days) instead of the domestic one (3-7 days). The SOP was right, the model just picked the wrong line from it.

5. Errors we looked at

Router mistakes. 14 out of 390. Almost all of them are between intents that are very close to each other: get_refund vs check_refund_policy (5), get_invoice vs check_invoice (3), place_order vs delivery_options (3), and one each for set_up vs change_shipping_address and track_refund vs check_refund_policy. Most of these pairs point to the same SOP anyway, so only 4 of the 14 ended up with the wrong document. One answer was `cancel_account`, which isn't even in our intent list. It still passed as valid JSON, so we should probably add a check against the catalog when parsing.

Retrieval mistakes. In an earlier run we used the SOP title plus its first section as the search string. delivery_options and delivery_period kept pulling `order_tracking.md` instead of `shipping_delays.md`, because those two SOPs have almost the same Scope section (they were also the 2nd most similar pair in our EDA, 0.38). We switched to using the full SOP text and after that all 27 intents found their own SOP. The 4 retrieval misses left in V2 are all because of router mistakes.

Hallucinations that are still there. V1 and V2 both have 12 flagged answers. In V2, 7 of them have placeholders like `[phone number]` or `[support@example.com]`. These come from the generator and not from retrieval. They mostly show up for contact_customer_service, complaint and track_refund, where the customer is asking for a contact or a timeline that the SOP doesn't really have, so the model fills the gap with something generic.

A prompt change that made things worse. At one point we added a line to the RAG prompt saying to only give timeframes, amounts and contacts that are in the SOP and to never make up a phone

number or link. Hallucinations actually went up, to 7.4% for V1 and 7.2% for V2, and placeholders showed up in 12-14 answers. It looks like mentioning phone numbers and links made the small model more likely to write them. We went back to the shorter prompt for the final numbers in this report.

6. Limitations

- Our ground truth for the answers is the SOP itself, not a proper human-written answer. So ROUGE/BLEU mostly reward copying policy wording and can't really judge if the answer is helpful or correct.
- The hallucination check is rule based. It catches made-up numbers, placeholders, fake contacts and asking for passwords, but it misses wrong claims that have no number in them. It also can't tell when a real SOP number is used in the wrong place, like the 10-21 days in the demo. So the 3-4% is probably on the low side.
- Consistency is 100% mostly because we use greedy decoding. It shows the pipeline is deterministic, not that it would be stable with sampling.
- Only 13 SOPs, each one is a single chunk, and we mapped the 27 intents to them by hand. A few mappings are a bit of a stretch (newsletter_subscription to data_privacy for example), which limits how grounded some answers can be.
- The adversarial set is small, 38 rows, so one mistake changes accuracy by around 2.6 points.
- We used a 1.5B model so it would fit on a free T4 in 4-bit. A bigger model would likely follow the SOP better.

7. Recommendations and future work

1. Go with V2. It is equal or better than the other two on every metric and way better on routing and retrieval. The router is also cheap to run (around 40 new tokens per query) and its JSON output is easy to log.
2. Handle hallucinations after generation instead of in the prompt. Our prompt experiment showed the small model doesn't follow "don't do X" rules well. A simple filter that removes any number, link or contact that isn't in the retrieved SOP would be more reliable. Another option is a second LoRA adapter trained on SOP-grounded answers.
3. Check the router output against the intent list, and if it's not a known intent, fall back to naive retrieval or ask the customer a clarifying question.
4. Split SOPs by section and retrieve the top 2-3 chunks, so the model gets the exact paragraph it needs (for example domestic vs international delivery times) and not the whole policy.
5. Add the actual contact channels to the SOPs. A lot of the leftover hallucinations happen because the SOPs don't list any phone or email, so the model invents one.
6. Make better test data: a small set of human-written reference answers, a faithfulness score from a human or an LLM judge, and more hand-written hard cases for the adversarial set (multiple intents in one message, other languages, prompt injection).